Unit 1

Part A: Local Environment Setup

1. Install VS Code — download from https://www.anaconda.com/download/success and install the Python extension from the Extensions Marketplace.
2. Install Miniconda — download from https://www.anaconda.com/download/ for your OS.
3. Create your course environment by running in your terminal:

```
conda create -n cc207 python=3.9
  conda activate cc207
```

4. Install PyTorch (CPU version):

```
pip install torch
```

5. Verify the install by running this in a new .py file or a Python shell:

```
import torch
 print(torch.__version__)
 print(torch.randn(3, 4))
```

Part B: Tensor Operations Lab

Create a file named tensor_lab.py in VS Code and complete the following tasks.
For each task, print both the shape and the values of the resulting tensor.

Task 1 — Dimensionality Create four tensors representing 0D (scalar), 1D (vector), 2D (matrix), and 3D (cube) data. Label each with a comment.

Task 2 — Broadcasting

```
x = torch.tensor([1.0, 2, 4, 8])
y = torch.tensor([2, 2, 2, 2])
```

Compute and print x + y, x - y, x * y, and x ** y. Predict each result on paper before running the code, then compare.

Task 3 — Broadcasting with mismatched shapes

```
a = torch.tensor([[1., 2.], [3., 4.]])   # shape (2,2)
b = torch.tensor([10., 20.])             # shape (2,)
```

Compute a + b. In a code comment, explain *why* this works even though the shapes differ (i.e., how PyTorch broadcasts b across a).

Task 4 — Reshaping Given tensor_3d below, reshape it into a (2, 4) tensor and then flatten it into a 1D tensor.

```
tensor_3d = torch.tensor([[[1., 2.], [3., 4.]],
                          [[5., 6.], [7., 8.]]])
```

Part C: Reflection

Answer in a short paragraph at the bottom of your script (as a comment) or in a separate text file:
In traditional IT, "Rules + Data = Answers." In Machine Learning, "Answers + Data = Rules." Using one real-world example (not from the lecture), explain what this shift means and why tensors are the data structure that makes it possible.

Submission Checklist

- Screenshot: conda environment + torch.randn(3, 4) output
- tensor_lab.py file with Tasks 1–4 completed and commented
- Terminal screenshot showing all tasks running successfully
- Reflection paragraph